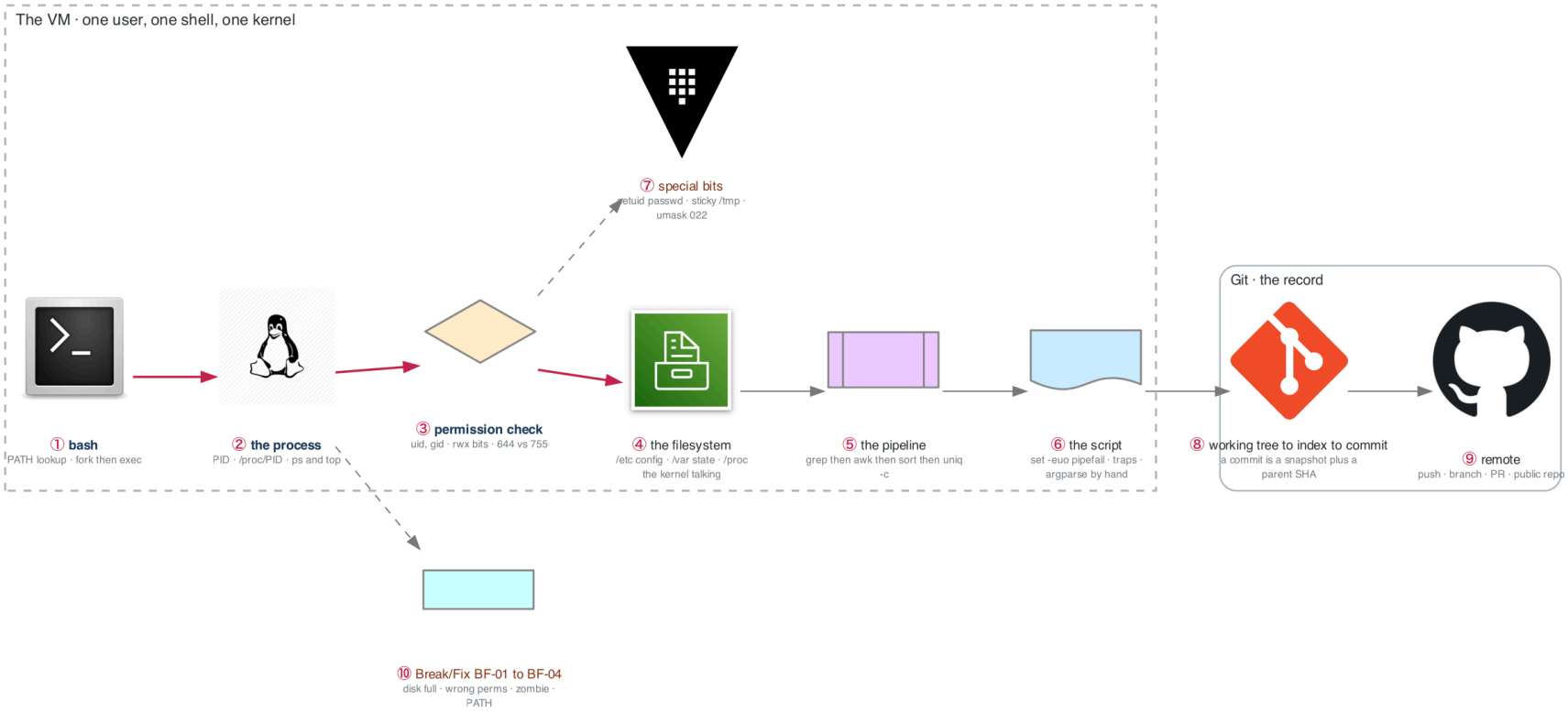
A process on a Linux system, from the shell to the commit, numbered

Week 1 · Figure 1.1 · figure with badges 1–10 · 2026-10-17



What to say, badge by badge

About three minutes at a normal pace (428 words). Point at each badge as you say its line.

Open
“This is one week of Linux in one picture: a command becomes a process, the kernel checks what it may touch, text flows through a pipeline, and Git records what you built. Follow the blue path first.”

1 bash
“Every command starts here. The shell searches PATH, forks itself, and the child execs the program. That is why a typo in PATH breaks everything at once, and why `which` is the first debugging command.”

2 the process
“The result is a process with a PID. /proc/PID is not a file on disk; it is the kernel describing that process. `ps` and `top` read it. When a lab says inspect a process, this is the directory to open.”

3 permission check
“Before any read or write, the kernel compares the file's owner, group, and mode bits to the process's uid and gid. 644 means owner writes, everyone reads. On a directory, execute means enter, which is why 755 matters there.”

4 the filesystem
“The hierarchy is a convention worth memorizing: /etc is configuration, /var is state that grows, /proc and /sys are the kernel. Where a file lives tells you who owns the problem.”

5 the pipeline
“Text processing is processes connected by pipes. grep selects lines, awk picks fields and counts into an associative array, sort and uniq -c aggregate. A 100,000-line log becomes a top-10 report with four processes.”

6 the script
“A real script starts with `set -euo pipefail`: exit on error, error on an unset variable, fail the pipeline if any stage fails. Traps clean up. Arguments are parsed by hand. This is `sysreport.sh`.”

7 special bits
“Three bits that are classic interview and escalation material. setuid lets `passwd` edit /etc/shadow as root. The sticky bit lets everyone write /tmp but delete only their own files. umask 022 is why new files are 644, not 666.”

8 working tree to index to commit
“A commit is not a diff. It is a snapshot plus the SHA of its parent, which is why history is a chain and why rewriting one commit changes every SHA after it. The index is the staging step in between.”

9 remote
“push sends the chain. A branch is a pointer, a pull request is a conversation about a pointer. The deliverable is public, which means the README has to work for a stranger.”

10 Break/Fix
“Four scenarios, forty minutes, narrated out loud: disk full, wrong permissions, a zombie, a broken PATH. Each one is logged in the incident journal. That journal is the interview story bank.”

Close
“So: shell to process to permission check to filesystem, text through pipes, a script that fails loudly, and Git recording it. Nothing in the next 25 weeks is not one of these in a costume.”